

Portfolio

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This repository contains a collection of professional and academic projects, including augmented reality research, UX case studies, and interactive prototypes. It serves as an archive of work completed between 2019 and 2025, spanning roles in industry, academia, and personal innovation.

Daniel Delgado, PhD – Professional Project Archive

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About

Daniel Delgado is a user experience researcher residing in Oceanside, California, with over seven years of experience designing and developing systems to enhance human-computer interaction. He earned his Bachelor of Science in Computer Engineering from Florida State University in 2019 and his Doctor of Philosophy in Computer Science from the University of Florida in 2025.

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Publications

Most Recent Publications

1. When Is Self-Gaze Helpful? Examining Uni- vs Bi-directional Gaze Visualization in Collocated AR Tasks → [View Publication](#)
2. Understanding User Needs for Task Guidance Systems Through the Lens of Cooking → [View Publication](#)
3. Evaluation of shared-gaze visualizations for virtual assembly tasks → [View Publication](#)

See more... → [View Publications](#)

🌟 Projects

1. Shared-Gaze Visualization in AR

Tech: Unity, MRTK, C#, HoloLens 2, Eye-tracking

Summary:

Developed a real-time shared-gaze system for collaborative industrial tasks. Conducted **60+ study sessions**, resulting in publications at **IEEE VR 2024** (*Best Poster Honorable Mention*) and **IEEE ISMAR 2025** → [View Project Folder](#)

2. EMG-Based Human-Machine Interface in Virtual Reality

Tech: Unity, C#, HTC Vive Pro 2 Virtual Reality **Summary:**

Designed and evaluated a VR-based training tool for EMG-controlled prosthetic hands. Conducted usability testing and performance analysis with **100+ hours** of user trials.

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3. Augmented Reality Task Guidance

Tech: Unity, HoloLens 2 **Summary:** Developed a multi-modal augmented reality system to guide users through complex cooking tasks while adpating to their needs. Additionally, provided a set of design guidelines and recommendations for future systems.

4. Multi-Modal Affect Detection

Tech: TensorFlow, CUDA, Python, High-Performance Computing **Summary:** Designed multi-modal convolutional neural network machine learning algorithms trained on audio/visual data to detect and classify real-time user facial expressions and emotions. Also developed methods for efficient feature extraction. → [View Project Folder](#) ____

Full Project List

→ [View Full Project List](#) ____

 Projects

Project	Description	Skills / Tech
Shared-Gaze Visualization in AR	Full-stack AR system for collaborative industrial tasks	Unity, MRTK, C#, Blender, Mixed Reality, UX Research
EMG-Based Human-Machine Interaction	VR-based training for EMG prosthetic hands	Unity, C#, C++, C, Python, Matlab, VR Interaction Design, Delsys
Augmented Reality Task	Multi-modal AR system for guiding users through complex tasks	Unity, C#, Contextual Inquiry, Qualitative Analysis

Guidance		
Affect Detection	Machine learning tools for analyzing user emotion through multi-modal techniques	Python, TensorFlow, Audio/Video Dataset

Skills & Tools

Research: UX Research, Eye-tracking Studies, Usability Testing, Experimental Design

Development: Unity, MRTK, C#, VR/AR Interaction Design

Design: Figma, Prototyping, Wireframing, Usability Testing **Data Analysis:** R, Python, RStudio, Matlab, Statistical Analysis, ArtANOVA **Collaboration:** Git

Resume

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Curriculum Vitae

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Teaching Materials

→ [View Teaching Materials](#)

Contact

 **Email:** danieldelgado.xr@gmail.com  **LinkedIn:** <https://www.linkedin.com/in/danieldel1996/>

Portfolio is maintained by **danielde196**.

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